SQL Handbook (51–100 Commands with Before & After Tables - Exact Model)

51 DISTINCT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT DISTINCT city FROM students;

After:
| Chennai |
| Madurai |
| Coimbatore |
```

52 LIKE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM students WHERE name LIKE 'R%';

After:
| Ravi |
```

53 IN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM students WHERE city IN ('Chennai','Madurai');

After:
| Ravi |
| Kumar |
```

54 BETWEEN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM students WHERE age BETWEEN 19 AND 21;

After:
| Ravi |
| Kumar |
| Anu |
```

55 ORDER BY DESC

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students ORDER BY age DESC;
```

```
After:
| Kumar | 21 |
| Ravi | 20 |
| Anu | 19 |
```

56 GROUP BY

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT city, COUNT(*) FROM students GROUP BY city;
```

```
After:
| Chennai | 1 |
| Madurai | 1 |
| Coimbatore | 1 |
```

57 HAVING

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT city, COUNT(*) FROM students GROUP BY city HAVING COUNT(*) > 0;
```

```
After:
All cities returned
```

58 COUNT DISTINCT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT COUNT(DISTINCT city) FROM students;
```

```
After:
3
```

59 SUM

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT SUM(age) FROM students;
```

```
After:
60
```

60 AVG

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT AVG(age) FROM students;
```

```
After:
20
```

61 MAX

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT MAX(age) FROM students;
```

```
After:
21
```

62 MIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT MIN(age) FROM students;
```

```
After:
19
```

63 LENGTH

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT LENGTH(name) FROM students;
```

```
After:
4,5,3
```

64 UPPER

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT UPPER(name) FROM students;
```

```
After:
RAVI, KUMAR, ANU
```

65 LOWER

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT LOWER(name) FROM students;
```

```
After:
ravi, kumar, anu
```

66 CONCAT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT CONCAT(name,'-',city) FROM students;
```

```
After:
Ravi-Chennai ...
```

67 SUBSTRING

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT SUBSTRING(name,1,2) FROM students;
```

```
After:
Ra, Ku, An
```

68 REPLACE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT REPLACE(name,'a','@') FROM students;
```

```
After:
```

R@vi, Kum@r, Anu

69 TRIM

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT TRIM(' Ravi ');

After:
Ravi

70 ROUND

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT ROUND(20.567,2);

After:
20.57

71 FLOOR

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT FLOOR(20.9);

After:
20

72 CEIL

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT CEIL(20.1);

After:
21

73 RAND

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

```
SQL:
SELECT RAND();

After:
Random number
```

74 NOW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT NOW();

After:
Current datetime
```

75 CURDATE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT CURDATE();

After:
Current date
```

76 DATE_ADD

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT DATE_ADD(CURDATE(), INTERVAL 1 DAY);

After:
Tomorrow date
```

77 DATEDIFF

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT DATEDIFF('2026-05-10','2026-05-01');

After:
9
```

78 EXTRACT

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT EXTRACT(YEAR FROM NOW());
```

```
After:
2026
```

79 JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students s JOIN courses c ON s.id=c.id;
```

```
After:
Joined rows
```

80 LEFT JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students s LEFT JOIN courses c ON s.id=c.id;
```

```
After:
All students
```

81 RIGHT JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students s RIGHT JOIN courses c ON s.id=c.id;
```

```
After:
All courses
```

82 CROSS JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students CROSS JOIN courses;
```

```
After:
All combinations
```

83 SELF JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT A.name,B.name FROM students A, students B WHERE A.city=B.city;
```

```
After:
Matching city pairs
```

84 UNION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name FROM students UNION SELECT name FROM teachers;
```

```
After:
Unique names
```

85 UNION ALL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name FROM students UNION ALL SELECT name FROM teachers;
```

```
After:
All names
```

86 CASE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name, CASE WHEN age>20 THEN 'Senior' ELSE 'Junior' END FROM students;
```

```
After:
Category output
```

87 SUBQUERY

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE age>(SELECT AVG(age) FROM students);
```

```
After:
```

```
Above avg
```

88 EXISTS

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM students WHERE EXISTS(SELECT 1);

After:
All rows
```

89 NOT EXISTS

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM students WHERE NOT EXISTS(SELECT 1 WHERE 1=0);

After:
All rows
```

90 VIEW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
CREATE VIEW v_students AS SELECT name FROM students;

After:
View created
```

91 SELECT VIEW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT * FROM v_students;

After:
Names list
```

92 DROP VIEW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
DROP VIEW v_students;
```

```
After:
View removed
```

93 INDEX

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
CREATE INDEX idx_name ON students(name);
```

```
After:
Index created
```

94 DROP INDEX

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
DROP INDEX idx_name ON students;
```

```
After:
Index removed
```

95 TRANSACTION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
START TRANSACTION;
```

```
After:
Transaction started
```

96 COMMIT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
COMMIT;
```

```
After:
Saved
```

97 ROLLBACK

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:
```
ROLLBACK;
```

After:
Undo

98 SAVEPOINT

Before Table (students):
```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:
```
SAVEPOINT sp1;
```

After:
Savepoint created

99 LOCK TABLE

Before Table (students):
```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:
```
LOCK TABLE students READ;
```

After:
Table locked

100 UNLOCK TABLE

Before Table (students):
```
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:
```
UNLOCK TABLES;
```

After:
Unlocked